Zecheng Zhu

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Research Interests

Reinforcement learning for whole-body humanoid manipulation: critic-only (value-based) RL, action chunking, and post-training of generalist vision-language-action policies; safe human–robot interaction; benchmarking learned and agent-developed policies.

Education

Imperial College London, MRes in Artificial Intelligence and Machine Learning — Oct 2026 – Sep 2027

- Started October 2026 with Prof. Stephen James in the Scalable Whole-body Intelligent Robot Learning (SWIRL) Lab, continuing the three projects below.

Imperial College London, BEng in Electronic and Information Engineering — Sep 2023 – Jun 2026

- Grade: 77.8/100 (First Class Honours), Top 5% of cohort
- Coursework: Robotics, Machine Learning, Deep Learning, Control Systems, Computer Vision

Publications

SafeYield: A Generalizable Recovery Framework for Safe Robot Manipulation
Zecheng Zhu, Stephen James. *Under review.*

BiGym 2.0: Benchmarking Learned and Agent-Developed Policies for Humanoid Household Manipulation
Zexi Zhang*, **Zecheng Zhu***, Zidong Chen, Zulkhuu Tuya, Stephen James. *Under review.*

Factored Q-Chunking
Zecheng Zhu*, Hongze Yu*, Stephen James. *Under review.*

*Equal contribution.

Research Experience

Imperial College London, Scalable Whole-body Intelligent Robot Learning (SWIRL) Lab — Sep 2025 – Present
Advisor: Prof. Stephen James

Factored Q-Chunking (FQC). Co-first author. Under review.

- In RL from demonstrations, Q-chunking commits every joint to one action chunk scored by one critic, though a base carrying two arms, or an arm carrying a dexterous hand, acts on different time scales. Co-developed FQC: the slow part commits for the full chunk, the fast part re-plans halfway, and each part's critic scores exactly its executed actions.
- Same actor and budget as Q-chunking: online fine-tuning success $0.102 \rightarrow 0.327$ on an arm and hand (DexJoCo, about $3\times$) and $0.302 \rightarrow 0.480$ on a mobile base with two arms (BiGym).

SafeYield. BEng individual project. Under review.

- Learned manipulation policies collide with, or are pushed off task by, a person entering the workspace. Built SafeYield, one module on a frozen policy that detects the person, yields, returns to the pre-interruption pose, and restores the policy's state so the task resumes rather than restarts.
- 10 BiGym tasks: collisions $78.8\% \rightarrow 1.6\%$, success $57.6\% \rightarrow 75.6\%$. Real Unitree G1 cupboard opening: 33/50 interrupted trials succeed, 4/50 collide (diffusion alone: 0/10 and 9/10). Zero-shot transfer: no collisions on ManiSkill 3; RoboCasa contact $100\% \rightarrow 20\%$ at unchanged 45.3% success.

BiGym 2.0. Co-first author. Under review. Data pipeline, training and evaluation, and part of the writing.

- Household-manipulation benchmark for a walking humanoid: BiGym adapted to the Unitree G1, with 20 tasks, 60 human virtual-reality demonstrations per task, and one frozen whole-body controller shared by demonstrations, learned policies and coding agents.
- Under identical observations and actions, a cold-start coding agent averages 53% on nine tasks, within three points of diffusion policy and above every demo-driven RL baseline; classical robotics tools lift it to 76%. Multi-object transport and cross-workspace stacking remain open for all methods.

PsiBot.AI Mar 2026 – Aug 2026
Research Intern, concurrent with SWIRL

- Built a VLM critic for reinforcement-learning post-training of an in-house VLA on a real dexterous-hand task, folding cardboard shipping boxes. Stage-aware progress labels replaced manual reward annotation across 658 demonstrations, reaching 30% autonomous success and a 50% gain in throughput.
- Compared critic backbones (ViT, DINOv2, VLM) and value targets (MC, TD, Q-learning) on 8×H20 GPUs; on BiGym, post-trained $\pi_{0.5}$ with progress labels in place of a VLM critic and MC return.

University of California, Berkeley Jun 2025 – Oct 2025
Hybrid Robotics Lab, part time

- Trained one transformer locomotion policy spanning quadrupeds, bipeds, and humanoids (G1, H1, A1, Cassie) over procedurally varied body shapes.

Huazhong University of Science and Technology Mar 2025 – Jun 2025
Advisor: Prof. Bo Tao, part time

- Fine-tuned a Unitree G1 locomotion policy to remain standing under shoves of up to 50 N, and deployed it on the real robot.

Industry Experience

Embodied AI Engineer *Internship*, part time, Lightwheel AI, Shanghai Jul 2025 – Oct 2025

- Contributor to Lwlab, a large-scale robot simulation framework: removed its RoboCasa dependency, rebuilt scene and object placement, and migrated RL pipelines from Isaac Lab, adding Unitree G1, PandaOmron, teleoperation and custom scenes.
- Trained a PPO pick-and-place policy that deployed zero-shot on a real LeRobot arm, using phased rewards, relative actions and targeted domain randomisation.

Teaching

Undergraduate Teaching Assistant & Peer Tutor, Imperial College London 2025 – 2026

- Led Control Systems lab sessions (2nd-year drone control demonstrations); peer-tutored first-year mathematics in 1-on-1 and small-group sessions.

Awards

Department of Computing Corporate Partnership Programme Prize 2026

Dean's List, Department of Electrical and Electronic Engineering, Imperial College London 2024, 2025

4th place, RSS 2026 workshop challenge on post-training robot foundation models 2026

Best Robotic Project Prize, Imperial College London 2024 – 2025
Autonomous campus tour-guide robot: self-balancing base (dual-loop PID/LQR), TD3 trajectory following, YOLOv8 person tracking.

Technical Skills

Programming and ML: Python, C++, C, PyTorch, JAX (Flax, Optax), Weights & Biases, Git, LaTeX
Methods: PPO, TD3, offline-to-online RL (CQN, CQN-AS, IQL, AWR, Q-chunking), diffusion and flow-matching policies, VLA post-training
Simulation and robotics: MuJoCo, MJX, BiGym, ManiSkill 3, RoboCasa, Isaac Lab / Isaac Sim, Gazebo, ROS
Robot learning and hardware: LeRobot ($\pi_{0.5}$), Unitree G1 real-robot deployment (FAST-LIO2 LiDAR SLAM, YOLOv8 human detection, ONNX policy export)
Compute: parallel training and evaluation sweeps on multi-GPU Linux servers